GLOBAL BRANDS GROUP

TEAM PROJECT · REAL-TIME VOICE AI SUPPORT

Banking Voice AI Support Agent – AMINA

A voice agent “Amina” that answers phone calls in real time from the customer’s own data and the bank’s own policies, and transfers to a human when it should.

Real-time voice | Data in SQL | Policy PDFs (RAG) | Built on Azure

1 · PROBLEM STATEMENT

Every call needs two lookups, and callers wait

Customers phone the bank about fees, blocked cards and disputes. A person answers, checks the customer's data in one system, searches the bank's rules in another, and talks the whole time.

Long waits

Routine calls (blocked card, unknown fee, transfer status) queue behind urgent ones.

Two systems, one answer

The agent finds the transaction in one tool, then reads a policy document in another.

Answers vary

Two agents can read the same policy differently, so the same question gets different answers.

2 · SOLVE THE PROBLEM

A voice agent that reads both sources

Database: who is this customer?

- Accounts and balances
- Transactions and fees charged
- Card status and limits
- Past tickets and disputes

Real PDFs: what are the rules?

- Egyptian Fee schedules and card terms
- Disputes and error-resolution rules based on CBE
- Central Bank of Egypt rules on electronic transfers
- When to transfer to a human

EXAMPLE CALL

"Why was I charged a fee on the 3rd?"

The agent verifies the caller, finds the transaction, finds the fee rule, and explains it out loud in a few short sentences.

Requirements for a production voice agent

What it must do

- Answer phone or browser calls in real time, by voice
- Verify the caller before reading any customer data
- Answer from the customer's records and the PDFs, with sources
- Create a request or ticket only after spoken confirmation
- Transfer to a human with a summary, any time the caller asks
- Keep a transcript, summary and log of every call

How well it must do it

- Speed: first spoken reply within 1s at p95
- Availability: 100% a month, with a fallback when a service fails
- Security: verified caller, read-only data access, secrets in Key Vault
- Privacy: redact personal data before logging; retention agreed with legal
- Safety: no made-up facts; guards against prompt attacks
- Operations: one trace per call; cost per call minute tracked

3 · HIGH-LEVEL ARCHITECTURE

From the caller's voice to the answer, on Azure

Caller
A phone call, or the browser microphone.

Azure Communication Services
Phone number, call control, two-way audio over WebSocket.

Voice agent service
Azure Container Apps. Runs the call session and calls the agents as tools.

Voice model
Azure Voice Live: speech in and out

Specialist agents
Data, Knowledge, Action

Transfer to a human
The call is transferred to staff. The console shows the summary, customer record and live transcript.

Data and monitoring
PostgreSQL, AI Search and Blob Storage hold the data. Application Insights traces every call.

The agents and what each one does

Agent	Job	Tools
Voice	Talks with the caller, speaks short answers	Azure Voice Live model
Triage	Finds intent and mood, picks who works next	Small, fast model
Data	Reads this customer's accounts and cards	get_transactions, get_card
Knowledge	Finds policy text, cites file and page	Azure AI Search
Action	Creates a dispute or ticket after "yes"	create_dispute, create_ticket
Checker	Reviews actions, flags unsupported answers	Rules plus LLM review
Handoff	Writes the summary, transfers the call	ACS call transfer

The Voice agent calls the others as tools. Data and Knowledge run in parallel to keep replies fast.

What happens on every call

1
Connect
The call arrives and audio starts streaming. The caller is verified before any data is read.

2
Listen
Speech becomes text as it is spoken. The system detects when the caller stops or interrupts.

3
Triage and plan
Classify intent and mood. Choose which agents to call.

4
Gather
Data and Knowledge agents run in parallel. A short “let me check” covers the wait.

5
Speak
The reply uses retrieved facts only, in short spoken sentences, and starts playing while it is generated.

6
Check and act
Actions need a spoken “yes”. Unsure or upset: transfer to a human. Log the turn.

Making it feel like a conversation

Interruptions
When the caller talks over the agent, speech stops at once and a new turn starts.

End of speech
Detect when the caller has finished. Too early cuts them off; too late feels slow.

Waiting well
While a tool runs, say “Let me check that.” Time out slow tools and recover.

Spoken style
Short sentences, one idea at a time, numbers read slowly, key details repeated back.

Latency budget
Voice-to-voice delay = network + speech recognition + model + tool calls + speech output. Measure each stage.
Team target: 1s at p95.

Database schema

customers
customer_id PK
full_name
phone
segment
joined_at
national_id

accounts
account_id PK
customer_id FK
type
currency
balance
status

cards
card_id PK
account_id FK
card_type
last4
status
daily_limit

transactions
txn_id PK
account_id FK
posted_at
amount
merchant
fee_code FK

fee_rules
fee_code PK
name
amount
applies_to
pdf_file
pdf_page

disputes
dispute_id PK
txn_id FK
opened_at
reason
status
resolution

tickets
ticket_id PK
customer_id FK
call_id FK
intent
status
summary

calls
call_id PK
customer_id FK
started_at
verified
sentiment
transcript_ref

Relations

customer → accounts → cards, transactions → disputes. customer → calls → tickets. transaction → fee_rules → PDF page.

KNOWLEDGE BASE

The real PDFs the agent reads

PDF	Type	Answers about
1stsource_personal_fee_schedule_2022.pdf	Fee schedule	Account fees
bankofireland_fees_2020.pdf	Fee schedule	Fees, second bank
cbc_personal_fees_2025.pdf	Fee schedule	Fees, recent
fabt_consumer_debit_mastercard_agreement.pdf	Card agreement	Card use, liability
teachersfcu_debit_dispute_form.pdf	Dispute form	Dispute steps
occ_bankwise_electronic_fund_transfer_act.pdf	Regulator guide	Disputes, deadlines
regulation_e_12cfr205_2020.pdf	US federal rule	Error resolution

Azure ingestion: PDFs in Blob Storage → Document Intelligence reads the layout → chunks keep file and page → embeddings → Azure AI Search index (keyword and vector).

Not covered yet: loans, KYC and fraud. Add real PDFs for them, or leave those questions out of scope.

WORKED EXAMPLE · SAMPLE VALUES

One call, start to finish

Caller says

"Why was I charged a fee on the 3rd?"

Behind the scenes

- Caller verified
- Data agent: transaction on the 3rd, fee_code [__]
- Knowledge agent: fee schedule, page [__]

Agent says

"On the 3rd there was a [amount] fee because your balance went below zero. It is on page [__] of the fee schedule. Shall I request a waiver?"

If the caller says yes, the Action agent creates a waiver ticket and a human decides. The agent never approves it.

Azure services and their roles

Need	Azure service	Role
Phone and call control	Azure Communication Services	Number, call control, two-way audio
Speech and voice	Azure Voice Live API (Speech)	Speech in and out, interruptions
Language model	Azure OpenAI in Foundry	Agent reasoning, embeddings
Policy search	Azure AI Search	Keyword and vector search of PDFs
PDF reading	Azure AI Document Intelligence	Layout-aware text from PDFs
Business data	Azure Database for PostgreSQL	Customer data, tickets, call logs
Files	Azure Blob Storage	PDFs, recordings, transcripts
Hosting	Azure Container Apps	WebSocket service that autoscales
Secrets and identity	Key Vault, Managed Identity, Entra ID	No keys in code
Monitoring and safety	App Insights, Monitor, Content Safety	Traces, alerts, guardrails

Optional: Azure Cache for Redis for call state, Azure AI Language for personal-data detection and sentiment.

Code, libraries and tools we build with

Layer	Choice	Why
Caller test client	React web page with microphone	Demo without a phone
Backend	Python, FastAPI, WebSockets	Streams audio and events
Azure libraries	Python SDKs for ACS, Voice Live, Search	Official, supported
Agent logic	Plain Python or LangGraph	Parallel agent calls
Data access	PostgreSQL with parameterised queries	Read-only, per customer
Local development	Docker Compose, dev tunnel	Call webhooks reach laptops
Delivery	GitHub Actions, Bicep, Container Registry	Repeatable deploys
Testing	pytest, k6 or Locust, scripted test calls	Regression and load

5 · APP FEATURES

What the app can do

Live voice conversation

Call a phone number or talk in the browser. Replies are spoken in real time.

Natural interruptions

The caller can talk over the agent, and it stops and listens.

Caller verification

No customer data is read until the caller passes the check.

Answers from data and PDFs

Account facts from the database, rules from the PDFs, with file and page.

Actions after “yes”

Disputes and tickets are created only after spoken confirmation.

Transfer to a human

The call moves to staff with a summary and live transcript.

Also: transcript, call summary and a tracing dashboard for the team.

6 · USER WORKFLOW

The caller’s path, and the transfer

1 · Call in
Dial the number, or press talk in the browser.

2 · Verify
Say your National ID and your birth date

3 · Talk
Ask in plain words. Interrupt any time.

4 · Confirm
Say yes before any request is created.

5 · Done
Resolved, or transferred to staff.

If the agent transfers the call

The caller is told a person is joining. The call is transferred to a staff member whose console shows the summary, intent and mood, customer record and live transcript, so the caller does not repeat anything.

PRODUCTION READINESS

Reliable, secure, observable

Reliability

- Time-outs and retries on every tool call
- If a service fails, say so and transfer to a human
- Autoscale on concurrent calls
- Fallback speech path if the real time path is down

Security and privacy

- Verify the caller first
- Read-only, per-customer data access
- Keys in Key Vault, managed identity
- Redact personal data before logs and traces
- Recording consent and retention set with legal

Observability

- One trace per call across all agents
- Latency per stage, interruptions, tool errors, transfer reasons
- Alerts on error rate and p95 latency

Before go-live: staff pilot first, then a few customers, with a human always one step away.

PRODUCTION READINESS

How we test it before callers do

Scripted calls

Seeded cases from the database with known correct answers, run after every change.

Grounding checks

Every fact in a reply must match a tool result or a PDF page.

Voice quality

Speech recognition on noisy and accented audio. Interruption detection and false stops.

Load tests

Many concurrent calls. Measure p95 and p99 latency and service limits.

Red team

Prompt attacks by voice or hidden in documents, and callers pretending to be someone else.

Staff pilot

Staff take real calls first. Review transcripts every week and fix what they find.

DATA AND REFERENCES

Where the data comes from

Source	Use in this project	Limit
Synthetic bank (we generate)	The SQL database	Fake, but known answers
Real public PDFs	The knowledge base	Check each source's terms
Test calls (we record)	Voice testing	Vary accents and noise
Banking77	Bank question wording	Labels only, no answers
Bank complaints dataset	Complaint wording	Confirm origin and licence
Bitext support dataset	Question phrasings	Synthetic

Azure: Voice Live API · ACS audio streaming · ACS with Azure OpenAI sample · Document Layout skill · Foundry tracing

Voice in production: Deepgram guide · Handoff pattern

Data: Bitext · Complaints data · Free call-center datasets